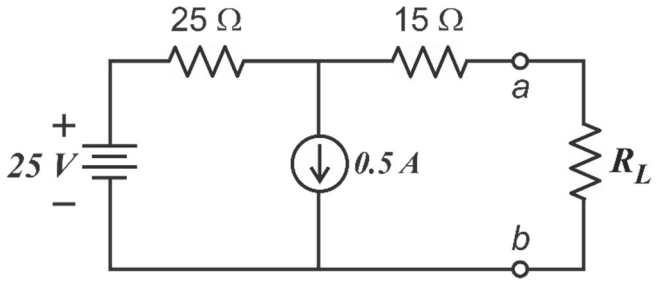
1) **Thévenin Equivalent Circuit**:
   A) Find the Thévenin Equivalent Circuit to the left of terminals ***a-b*** for the following circuit.
   B) Using the Thévenin Equivalent Circuit, calculate the power delivered to the load resistor when the load resistor (R_L) equals the following three cases:
      i. 100 Ω
      ii. 40 Ω
      iii. 3 Ω
   C) For the three cases in (B), at what R_L does the "maximum power to the load" occur? Can you make a statement about this condition?



2) **Norton Equivalent Circuit:** Using the circuit in Problem 1, find the Norton Equivalent Circuit to the left of terminals ***a-b***. **Sketch the equivalent circuit** including R_L. You do not need to calculate the power to the load for this circuit.

3) **Source Transformation**: Using the circuit in Problem 1, find the equivalent circuit using Source Transformation to the left of terminals ***a-b***. You will need to apply two steps of Source Transformation, **Sketch the equivalent circuit** including R_L. You do not need to calculate the power to the load for this circuit.

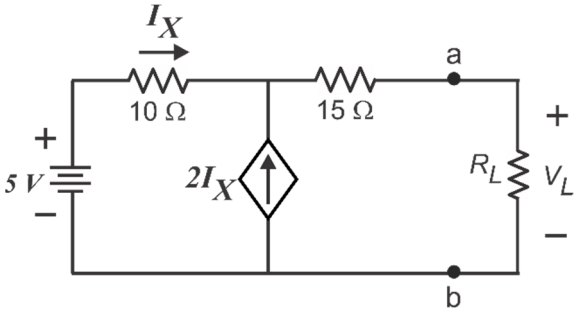
4) **$R_{TH} = R_N$:** Verify that the Equivalent Resistance, $\boldsymbol{R_{EQ}}$, found in the Thevenin and Norton solutions above can also be calculated by the ratio of $\boldsymbol{V_{TH}}$ (problem 1) and $\boldsymbol{I_N}$ (problem 2).

$$\boldsymbol{R_{EQ} = R_{TH} = R_N = \frac{V_{OC}}{I_{SC}} = \frac{V_{TH}}{I_N}}$$

5) **Thévenin Equivalent Circuit including a Dependent Source**

   Find the Thevenin Equivalent Circuit by

   A) Calculating $\boldsymbol{R_{TH}}$ using V_{TH}/I_N
   B) Calculating $\boldsymbol{R_{TH}}$ using a test source connected to terminals a-b (remove the load). All independent sources from the original circuit at set to zero. Dependent sources remain in the test circuit. You may use a voltage source or a current source as the test source. For example, it is relatively easy to use a voltage source of 1 VDC as the test source. Find the associated test source current, I_{TS}, then calculate $\boldsymbol{R_{TH}} = 1V/I_{TS}$



6) **Source Transformation**: Using Source Transformation, simplify this circuit to the left of terminals a-b. You will need to use "two" steps of Source Transformation. Combine parallel/series circuit elements as needed. **Sketch the equivalent circuit** including R_L, provide values for $\boldsymbol{V_S}$ and $\boldsymbol{R_S}$.

